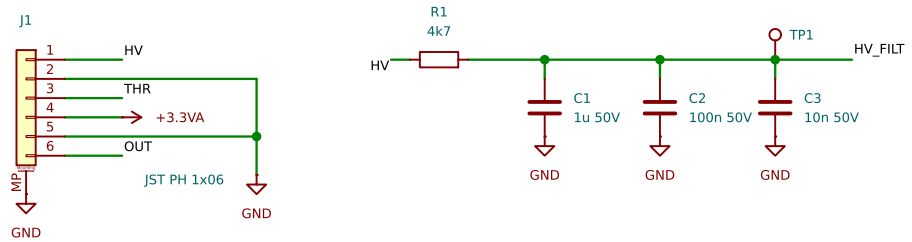
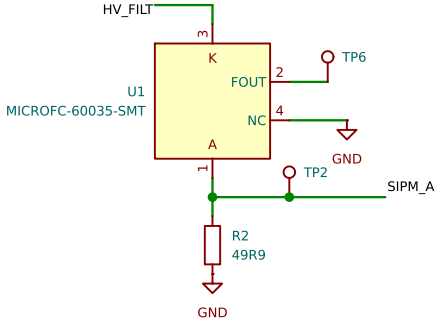
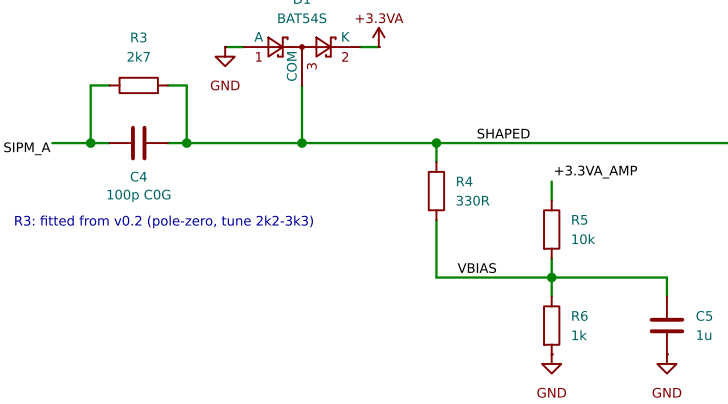
1. Connector and bias filter



2. SiPM and load

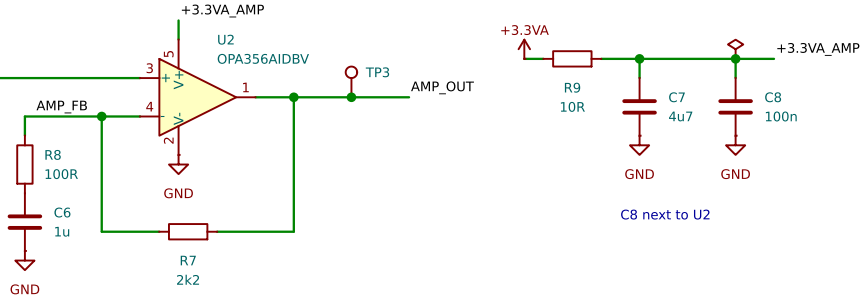


3. Shaping and baseline



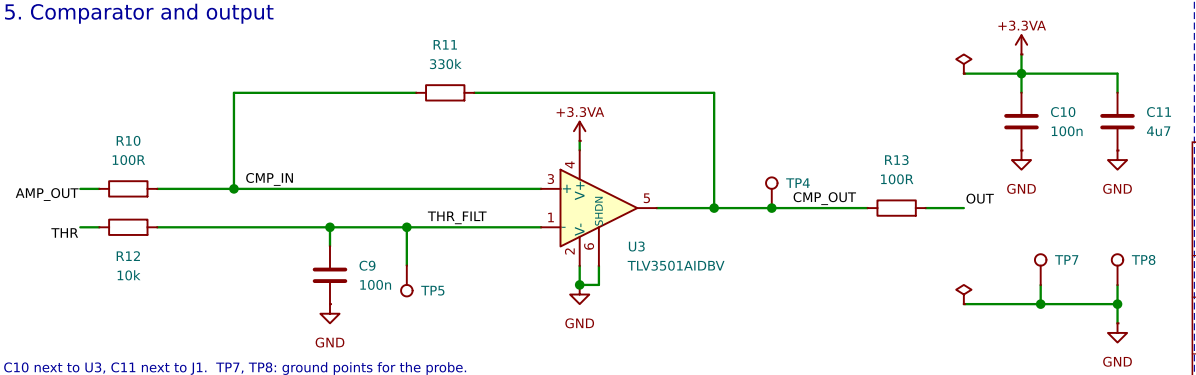
R3: fitted from v0.2 (pole-zero, tune 2k2-3k3)

4. Amplifier



C8 next to U2

5. Comparator and output



C10 next to U3, C11 next to J1. TP7, TP8: ground points for the probe.

PCB-401: one SiPM channel (bias filter, shaping, amplifier, comparator)
R2, C4 and R3 are starting values, to be tuned at bring-up

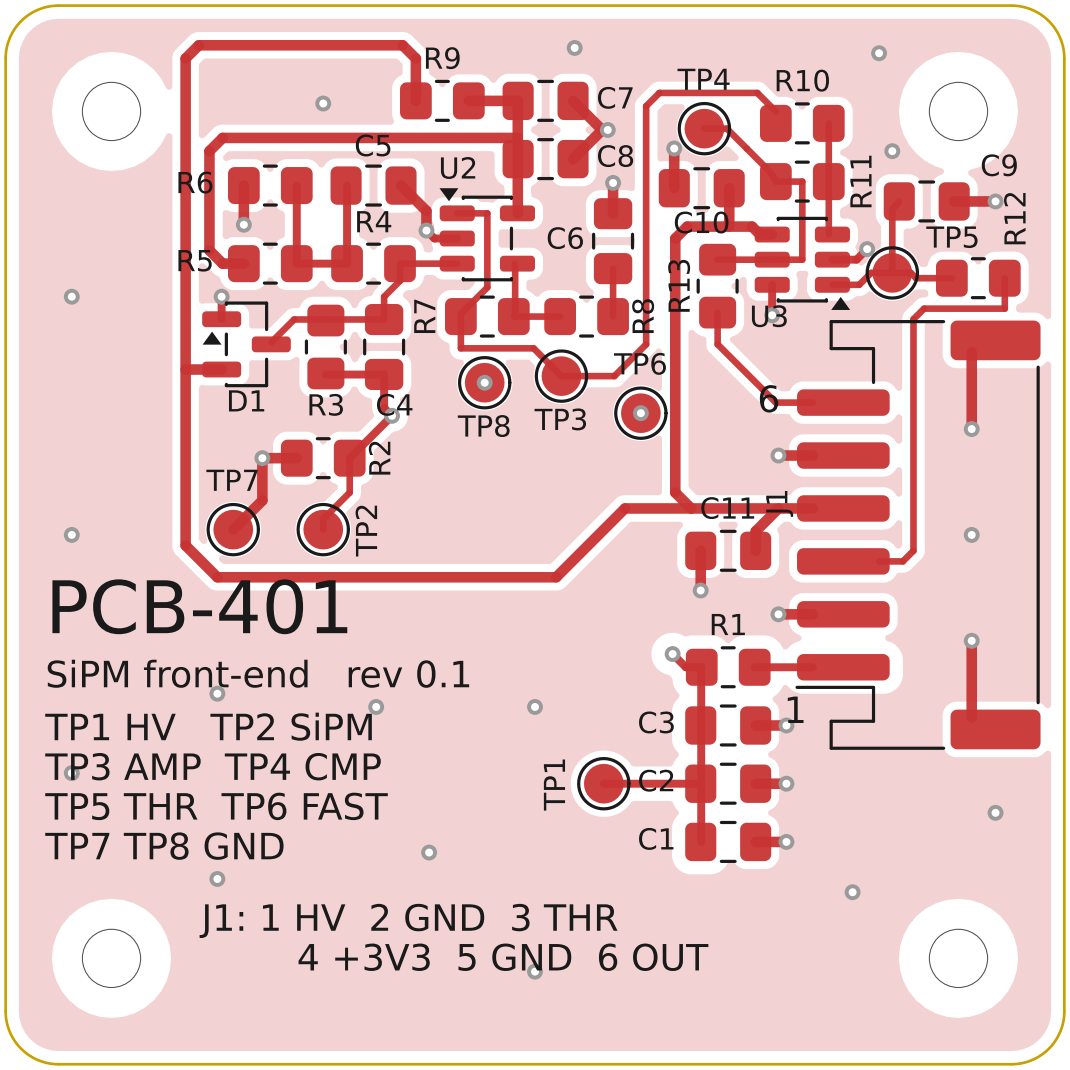
Muon time dilation project

Title: Front-end channel board

Size: A4 Date: 2026-10-12 Rev: 0.2 File: PCB-401_front-end_channel_board.kicad_sch

PCB-401: top view (copper, silkscreen, outline)

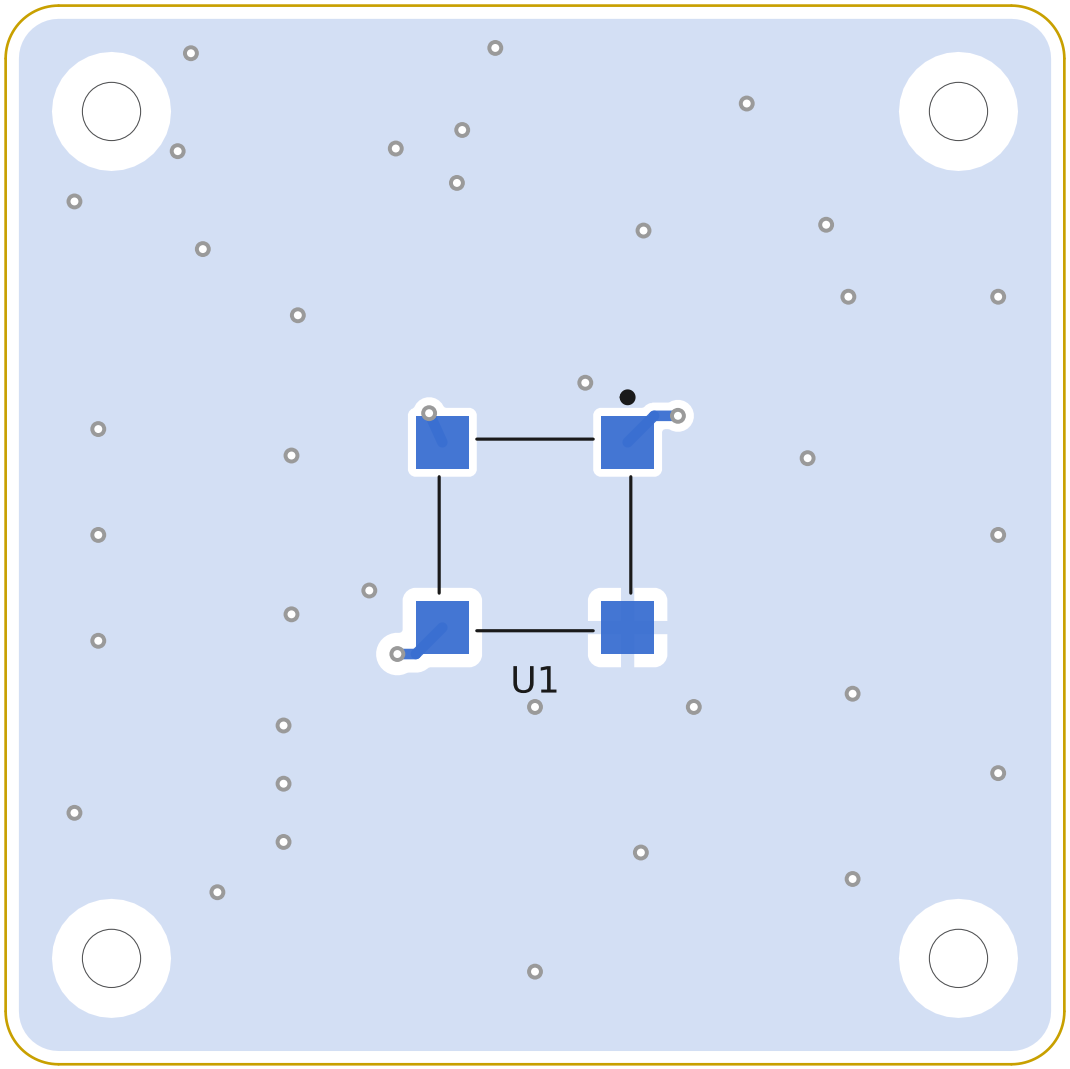
PCB-401_front-end_channel_board.kicad_pcb · board 40.0 x 40.0 mm · scale 3.5:1 on A4 · rev 0.1



10 mm

PCB-401: bottom view, mirrored (as seen when the board is flipped)

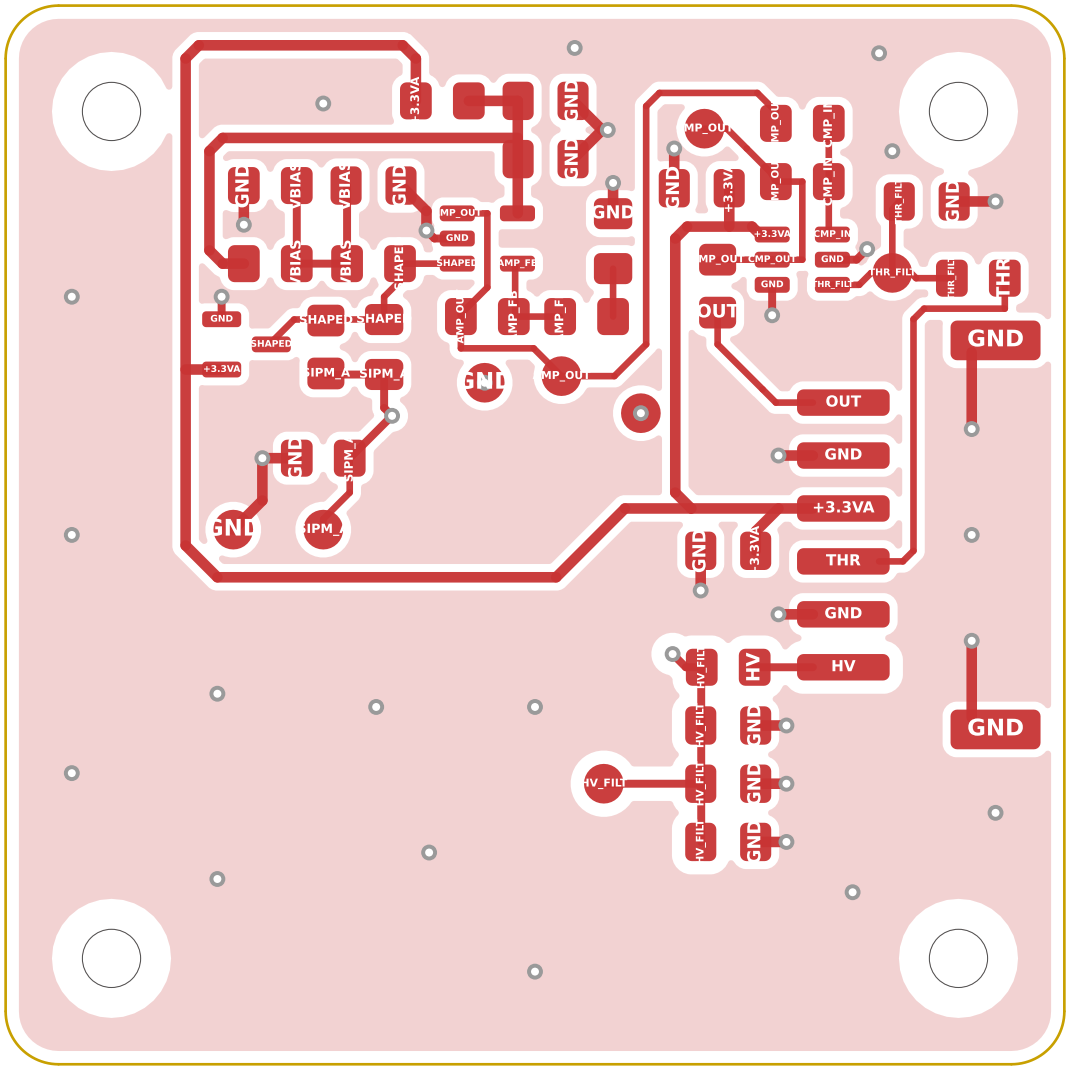
PCB-401_front-end_channel_board.kicad_pcb · board 40.0 x 40.0 mm · scale 3.5:1 on A4 · rev 0.1



10 mm

PCB-401: top copper (F.Cu) with net names on the pads

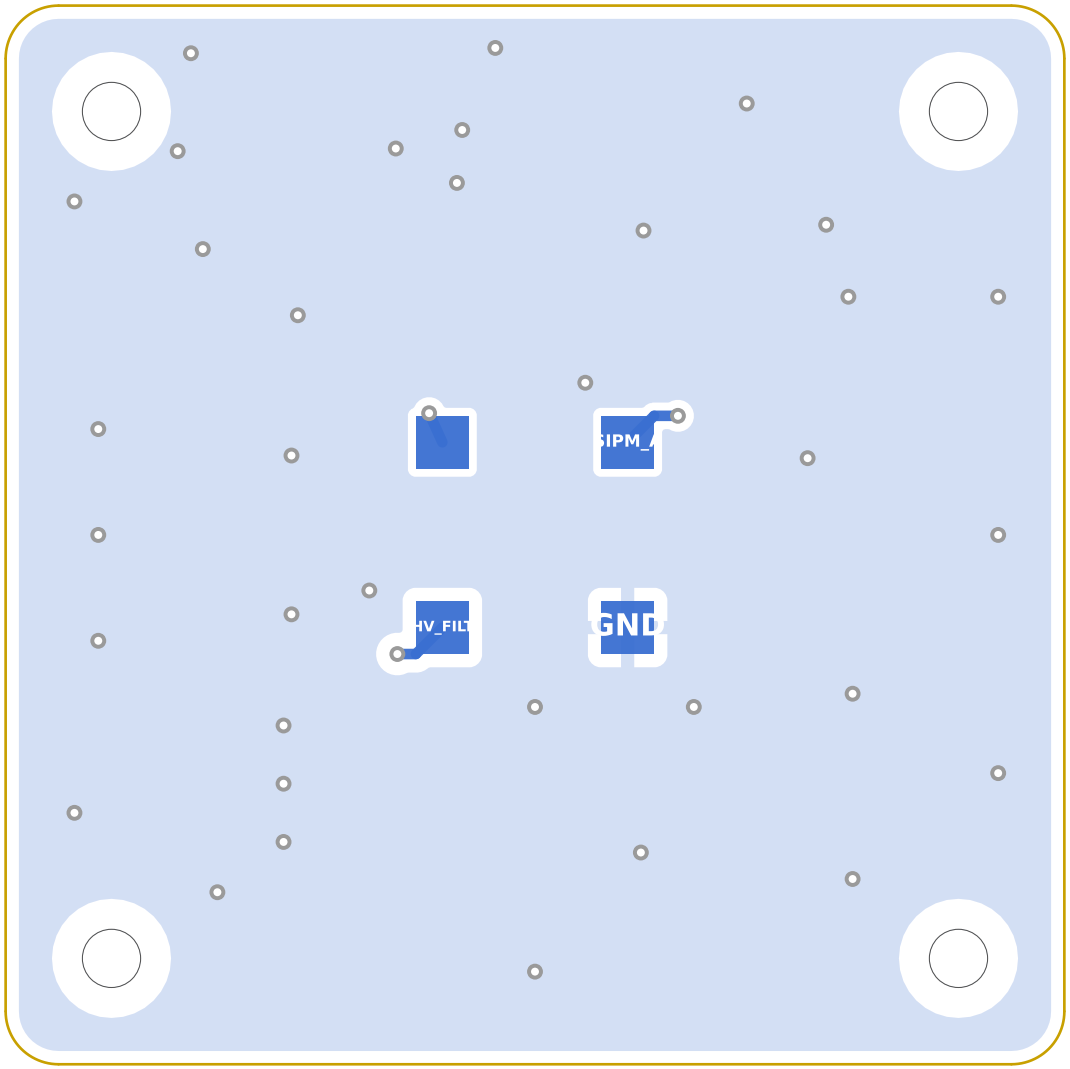
PCB-401_front-end_channel_board.kicad_pcb · board 40.0 x 40.0 mm · scale 3.5:1 on A4 · rev 0.1



10 mm

PCB-401: bottom copper (B.Cu), mirrored, with net names on the pads

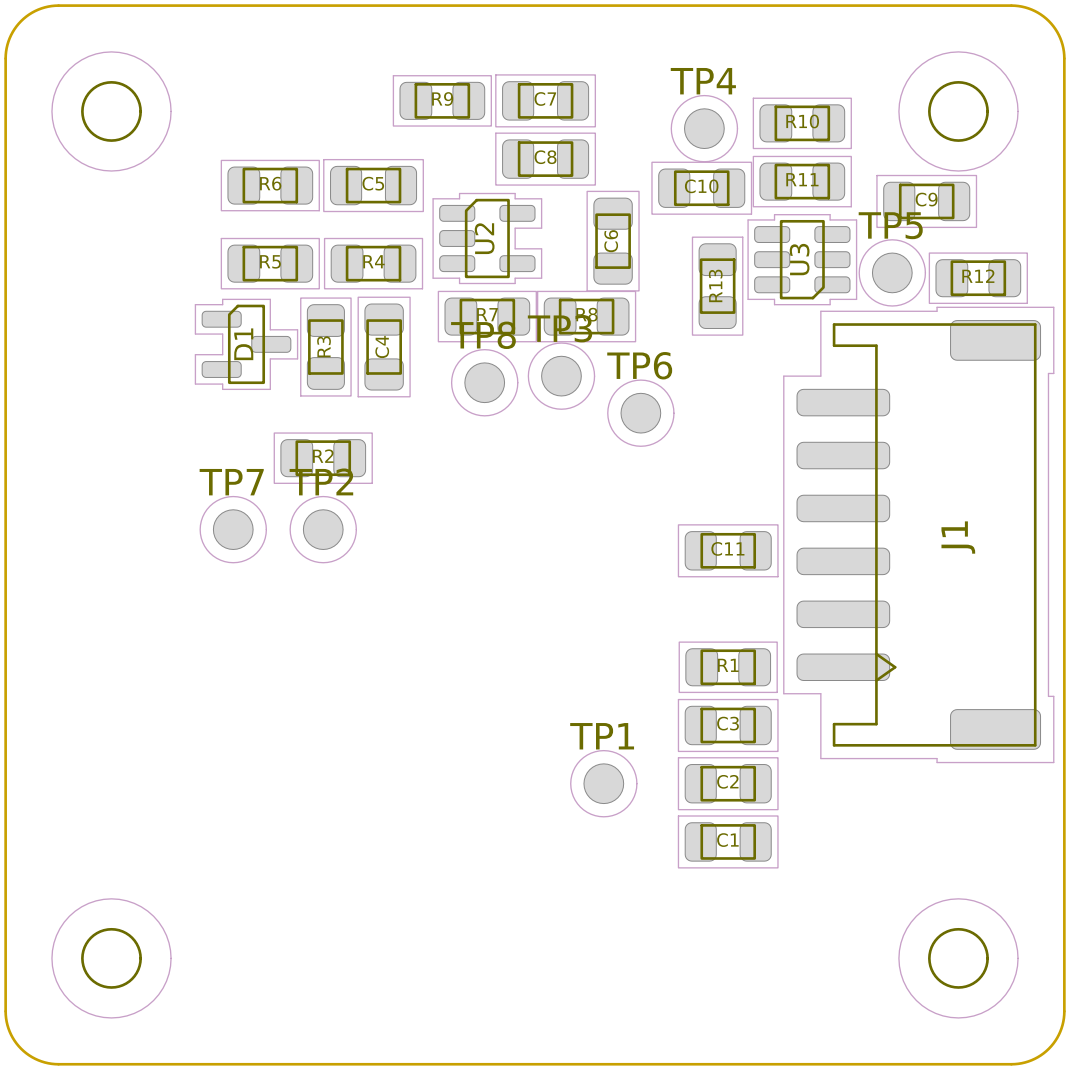
PCB-401_front-end_channel_board.kicad_pcb · board 40.0 x 40.0 mm · scale 3.5:1 on A4 · rev 0.1



10 mm

PCB-401: top-side assembly (references and component outlines)

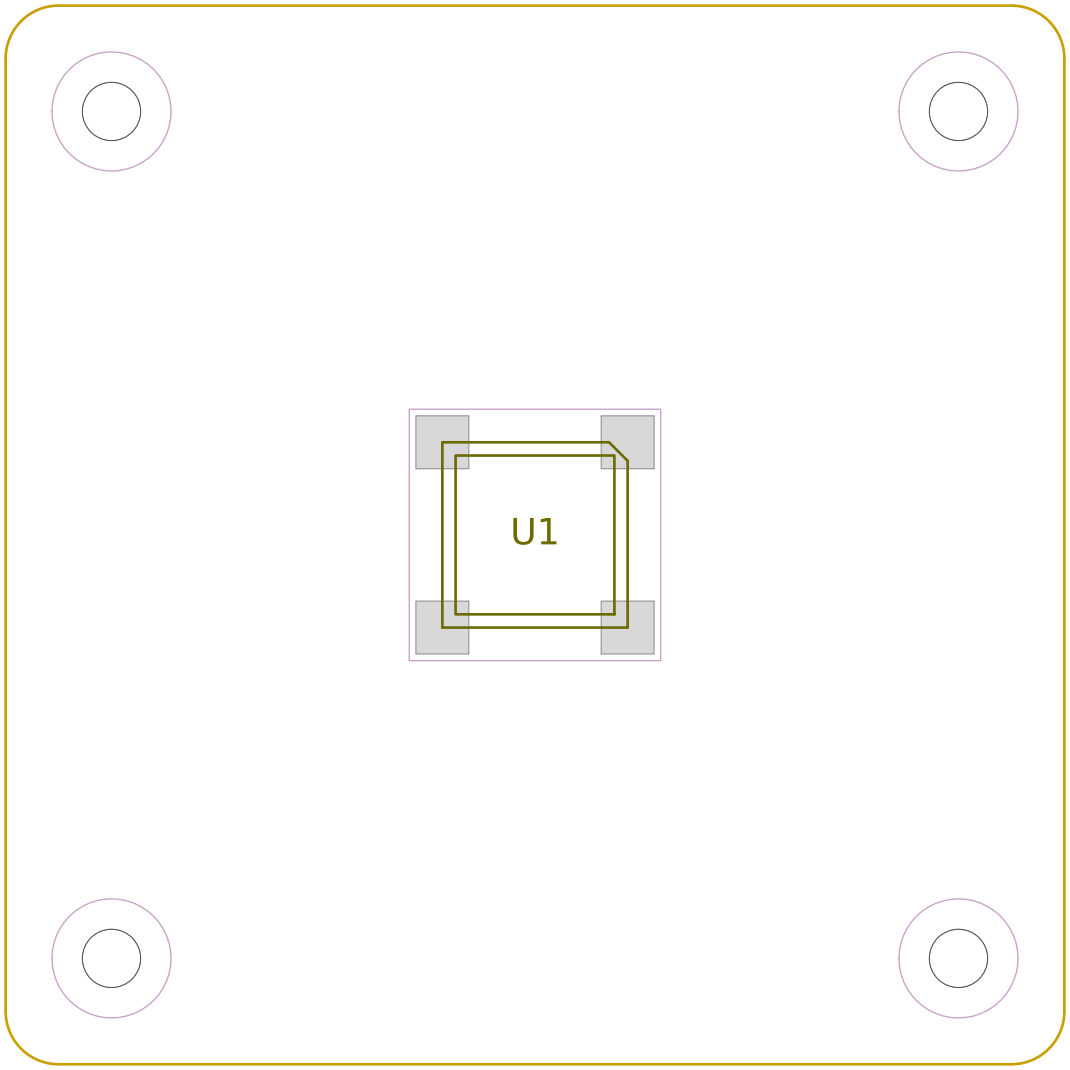
PCB-401_front-end_channel_board.kicad_pcb · board 40.0 x 40.0 mm · scale 3.5:1 on A4 · rev 0.1



10 mm

PCB-401: bottom-side assembly, mirrored

PCB-401_front-end_channel_board.kicad_pcb · board 40.0 x 40.0 mm · scale 3.5:1 on A4 · rev 0.1



10 mm